Prospectus Outline

Matthew

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Abstract

The mass 9 nuclear systems represents a significant deviation from the IMME, prompting investigation over non-trivial effects arising from isospin and nuclear structure considerations.

1 Introduction

Context.

One of the great triumphs of the past century of physics has been the illumination of the internal structure of the atom. It is well known that atomic nuclei are comprised of protons and neutrons and are held together by the strong force. The constituent nucleons of these nuclei arrange themselves within discrete shells quantized by angular momentum. These nuclear configurations are described by quantum numbers, such as n, l, and m for differing energy levels, angular momenta and multiplicity. Another quantum number used for characterizing nuclei with differing concentrations of protons and neutrons is isospin.

Isospin is a quantum number that is conserved for strong interactions. It is equivalent to spin for nucleons and takes the values $\mp\frac{1}{2}$ for protons and neutrons respectively. Isospin is derived from W. Heisenberg's attempt to treat both protons and neutrons as two states of the same particle, which was expanded upon by E. Wigner and given the name 'isotopic spin' [**?**]. In particular, this isotopic spin, or isospin, plays a crucial role in nuclear configuration characterization. An example are isobaric analog states (IAS), which are defined as states within the same isospin multiplet T with differing isospin z-projections T_z. Due to the presence of Coulomb interactions from proton-proton repulsion, the isospin symmetry between nuclei in isospin multiplets is broken. This gives rise to differences in energy levels between IAS [**?**]. page 26. The contributions to these energy shifts arise from various admixtures from interactions within the nucleus.

The Thomas-Ehrman shift is an energy shift between the energy states in mirror nuclei by the suppression of the proton-rich nucleus's energy levels relative to the proton-deficient nucleus. It was initially proposed by J. Ehrman and R. Thomas to describe the displacement in the first excited states between C^{13} and N^{13}. Due to the insufficiency of the two-body Coulomb repulsion to adequately explain the energy shift, Ehrman and Thomas employed the R-matrix formalism to investigate the effect of boundary conditions near the nuclear surface on energy level shifts. It was found that the application of this formalism resulted in a sufficient shift in energy levels between the mirror nuclei but was heavily dependent on the value of the nuclear radius involved [**?**][**?**]. From these findings, the characteristic shift between energy levels in mirror nuclei from the two-body interaction of a core and valence nucleon is referred to as the Thomas-Ehrman shift.

The nuclei ^9B and ^9Be are suitable candidates for this energy shift. It is widely thought that the displacement in energy levels between the nuclei are caused by this effect. However, this requires a precise measurement of energy levels between both nuclei, and while many states in ^9Be are well-known, the same cannot be said about ^9B. In particular, the $\frac{1}{2}+$ state has not been accurately measured in contrast to the corresponding state in ^9Be at 1.684 MeV. An accurate measurement of this state is necessary to validate the Thomas-Ehrman shift for these mirror nuclei.

The masses of nuclei within an isobaric multiplet are described by the Isobaric Multiplet Mass Equation (IMME):

$$M(T_z) = a(\alpha, T_z) + b(\alpha, T_z)T_z + c(\alpha, T_z)T_z^2$$

where α refers to any other relevant quantum numbers besides the isospin and the z-projection. This equation is derived by assuming the nucleons interact via the strong interaction, H_0, then assuming that the two-body Coulomb interactions between protons can be treated as a perturbation to H_0. This results in an equation that is quadratic in T_z, requiring at least $T = \frac{3}{2}$ isospin states to validate this equation. In order to deduce the coefficients to this equation, a fit to the mass excess for the nucleus in question is required, usually deduced from a measurement of Q-value [**?**]. The only known mass quartet to significantly deviate from this equation is the A = 9 $T = \frac{3}{2}$ quartet, involving ^9Li, ^9Be, ^9B, and ^9C.

The value of the energy of the first $\frac{1}{2}+$ state of ^9B is currently an unresolved issue within nuclear

physics. After the first attempted measurement of this state in 1955 [**?**], there have been many attempts to measure this state with various reaction channels. Unlike its mirror nucleus ^9Be, the ground state of ^9B is unbound with a proton separation energy of -185.9 keV, meaning that all excited states exist as resonances. It lies in a region containing many broad resonances, making the task of measurement challenging.

1.1 Reaction Kinematics

To populate states within nuclei, a commonly used method is to set up a reaction with known parameters that allows for the deduction of information from the populated nucleus. This involves a projectile nucleus, typically a light nucleus that has some energy with respect to the lab frame and a target nucleus that is at rest. The projectile interacts with the target, which results in a residual nucleus and an ejected nucleus, or ejectile, that travels outwards and is detected in spectroscopy. It is conventional to represent this type reaction as $A(a,b)B$, where a is the projectile, A is the target, B is the residual, and b is the ejectile. This reaction describes a transfer reaction, in which the interaction involves a transfer of nucleons between the projectile and target nucleus, and is the primary means used to populate states in ^9B for this project. In this case, the reaction is $^7Li(^3He,n)^9B$. In order to observe the residual nucleus created, a measurement of the resulting ejectile yields information based on kinematics. In our case, this ejectile is simply a neutron.

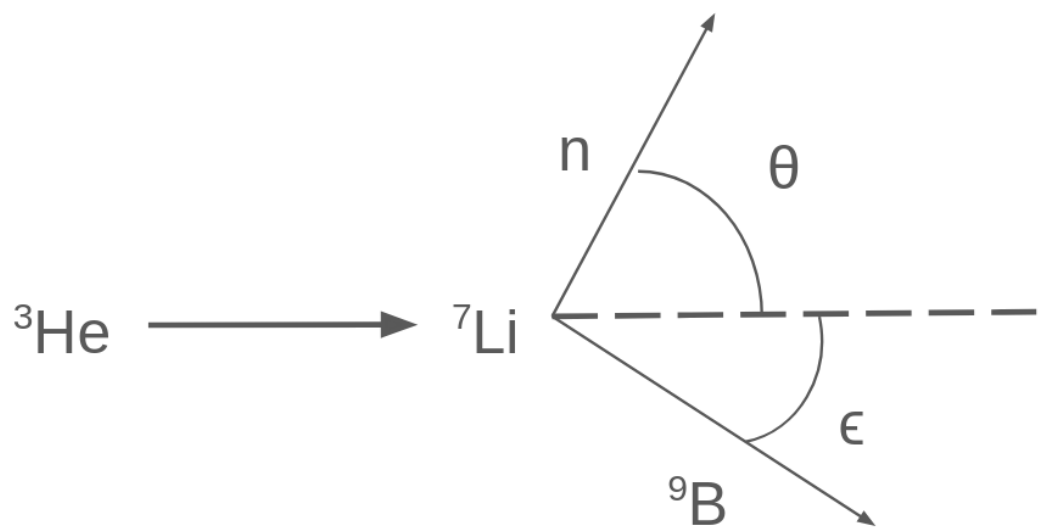

Figure 1: Kinematic representation of transfer reaction.

Neutron spectroscopy is the study of nuclear states from the measurement of neutron spectra. A technique for measuring this spectrum is time-of-flight (ToF) spectroscopy. ToF spectroscopy works by measuring the flight time of the neutron from the target to the detector. Since the neutron is a massive particle, its ToF is inversely proportional to its energy. In order to accurately determine the neutron flight time, the flight times of the prompt gammas emitted from excitations of the target from the beam are used. Since the gammas travel at the speed of light, a distinct peak can be identified within a ToF spectrum, allowing for the determination of neutron flight times. An advantage of this technique is the separation of neutrons from different excited states due to the dependence of ToF on neutron energy. This allows for the examination of all populated excitations of a given residual nucleus.

To derive the energies populated by the residual nucleus from the ejectile, one applies conservation of momentum p and energy E

$$p_a + p_A = p_b + p_B(1) E_a + E_A = E_b + E_B(2) \tag{1}$$

where $E = \frac{p^2}{2m} + mc^2$. The conservation of momentum can be written as

$$p_a = p_b cos\theta + p_B cos\epsilon(3)0 = p_b sin\theta + p_B sin\epsilon(4) \tag{2}$$

Using $p = \sqrt{2mT}$ with kinetic energy T and substituting (4) into (3), we have

$$T_B = \frac{1}{m_B}[m_a T_a + m_b T_b - 2\sqrt{m_a m_b T_a T_b} cos\theta$$

.

Going further, we can take the Q-value, or the difference in mass energy between the initial and final nuclei, which is given as

$$Q = (m_a + m_A - m_b - m_B)c^2$$

Using (1) and rearranging with $E = T + \frac{p^2}{2m}$,

$$Q = (m_a - m_b - m_c)c^2 = T_b - T_a + T_b = T_b - T_a + \frac{1}{m_B}[m_a T_a + m_b T_b - 2\sqrt{m_a m_b T_a T_b} cos\theta$$

The excitation of the residual nucleus can then be expressed as

$$Q = Q_0 - E_{ex}(5)$$

,

where Q_0 is the Q-value of the ground state in ^9B and E_{ex} is the excitation energy of the nucleus. The excited states of ^9B is described by

$$E_{ex} = Q_0 - (m_a + m_A - m_b - m_B)c^2 = Q_0 - T_b - T_a + T_B = Q_0 - T_b - T_a + \frac{1}{m_B}[m_a T_a + m_b T_b - 2\sqrt{m_a m_b T_a T_b} cos\theta])$$

.

Regarding neutron interaction with scintillators, the primary means of energy transfer is via elastic scattering. The maximum amount of energy transferred can be parameterized as $E' = \frac{A-1}{A+1}E$. For deuterium, $A = 2$, so $E_{dep} = \frac{8}{9}E$. From this energy, the light-output can be parameterized as [**?**]

$L = aE_{dep} - b(1 - e^{-cE_{dep}})$

where a, b, and c are parameters that must be experimentally characterized. These coefficients have been determined via previous work. The table below is adapted from [**?**].

States in nuclei are further characterized by their total spin J and their parity Π. This J^Π value features within angular distribution plots that plot the differential cross section $\frac{d\sigma}{d\Omega}$ over the center of mass angle θ_{cm}. The differential cross section is a measure of the likelihood of a given reaction occurring as a function of the angular distribution of the reaction products. In order to compute this angular distribution, we use the equation

$\frac{d\sigma}{d\Omega} = \frac{N}{\epsilon\rho I d\Omega}$

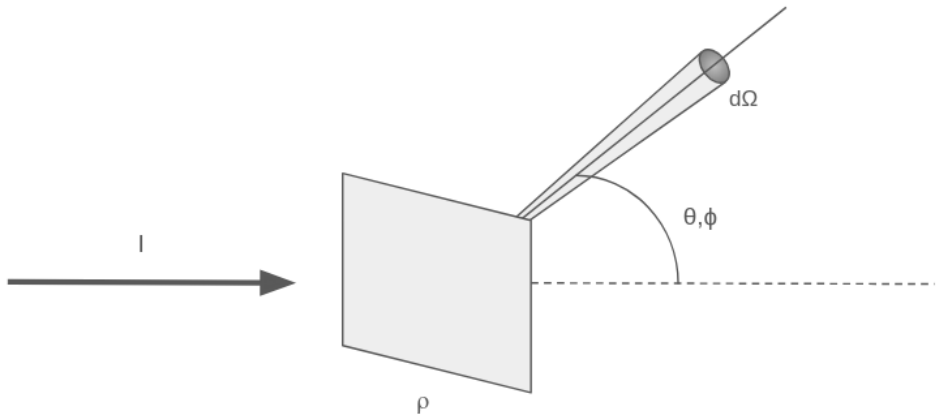

Figure 2: Visual example of kinematics in cross section calculation.

where N is the number of ejectile nuclei detected, ϵ is the detector efficiency, ρ is the atomic density of the target, I is the number of projectile nuclei incident to the target, $d\Omega$ is the solid angle of the detector. These experimentally determined angular distributions can then be compared to theoretical angular distributions using the Distorted Wave Born Approximation (DWBA) and optical model to find spectroscopic factors, denoted as C^2S. The experimental and theoretical angular distributions are related by

$(\frac{d\sigma}{d\Omega})_{experiment} = C^2S(\frac{d\sigma}{d\Omega})_{theory}$.

2 Experimental Procedures

2.1 Tools of the Trade

CATRiNA

In order to detect particles such as a neutron, scintillators are used. Within the field of nuclear physics, it is customary to distinguish the type of neutron radiation based on the energy range of the observed neutrons. Fast neutrons are typically considered to be greater than 1 MeV in kinetic energy. Since the cross section for neutron interactions in most materials is strongly dependent on neutron energy, fast neutron detection relies on elastic scattering interactions. Scintillators function as a reliable way to determine neutron energy spectra. Scintillators refer to any material that emits light after interacting with radiation. This light is then converted into an electric signal via a Photomultiplier Tube (PMT).

The Compound Array for Transfer Reactions in Nuclear Astrophysics (CATRiNA) is an array of deuterated benzene scintillators. The scintillators are stored in a cylindrical container with two sizes: 4"x2" and 2"x2". The 4"x2" and 2"x2" scintillator is the EJ-315 deuterated liquid scintillator produced by Eljen Technology. The coupled Photomultiplier Tube (PMT) is the 9821B PMT developed by ET Enterprises for the 4"x2" scintillator and the Hamamatsu R7724 PMT for the 2"x2" scintillator. Both scintillators rely on the scintillation of deuterated benzene to accurately measure incoming neutron energies. The scintillating molecule, benzene, comprises a hexagon of carbon atoms with a deuterium atom bonded to each carbon atom. About two or three of the valence electrons of carbon occupy the σ orbital in between the carbon and deuterium atoms and are strongly localized. The remaining electrons occupy the π orbitals, which extend outside of the molecular plane and are not strongly localized [**?**]. Transitions between the π orbital electrons are responsible for the luminescence seen within the deuterated benzene. A level diagram is provided in Figure. The electrons can be excited into either singlet or triplet states, denoted as S_i and T_i in Figure. Only transitions between levels emit gamma radiation. Transitions between vibrational states within the same level thermally de-excite. There are three main decays that can occur to excited π orbital electrons.

Prompt fluorescence refers to an electron, excited into a singlet state, decays into the ground state. This transition is short lived, with a half-life on the order of 1-80 ns. Phosphorescence refers to a process where a singlet electronic configuration is converted into a triplet configuration via inter-system crossing. Since the triplet has a much longer lifetime than the singlet with up to 10^{-3} s, this results in a delayed form of emission. Since the triplet state is at a lower energy relative to the singlet state, the emitted photon will have a longer wavelength than fluorescence photons. Delayed fluorescence refers to the process where a singlet state, converted into a triplet state, converts back into a singlet state before decaying via fluorescence [**?**]. The intensity of the fluorescence decays can be represented by the $I = Ae^{-t/\tau_f} + Be^{-t/\tau_t}$, where τ_s and τ_f refers to the mean lifetimes of the slow and fast components of the emission spectrum (delayed fluorescence and fluorescence) and A and B corresponds to the relative magnitude of each component.

These two decays are basis of the Pulse-Shape Discrimination (PSD) technique for separating n-γ events. The relative magnitude of each component of the emission spectrum is dependent upon the type of incident radiation. γ radiation typically has a strong prompt fluorescence component but a small delayed fluorescence component. Neutron radiation has a smaller prompt fluorescence component but a larger delayed fluorescence component. This allows for the discrimination of these events via long and short gates. As shown in Figure, the long gate is the total integral of the signal while the short gate is an integral over the rise time. The tail gate (long integral - short integral) can then be defined. This "tail integral" is consistently greater for neutron-induced events than γ-induced events, resulting in a PSD spectrum where each tail corresponds to a different radiation type.

Referring to the light-output incident radiation emits when interacting with scintillators, it is customary to define the unit electron-volt electron equivalent, or eVee. An eVee is defined as the light-output an electron with 1 ev of energy gives off. The PSD spectra obtained in this prospectus are plotted in terms of keVee and PSD ratio. The value of the long integral is defined in terms of arbitrary units, so a calibration is applied using ^{60}Co and ^{137}Cs. These two sources contain well-known Compton edges

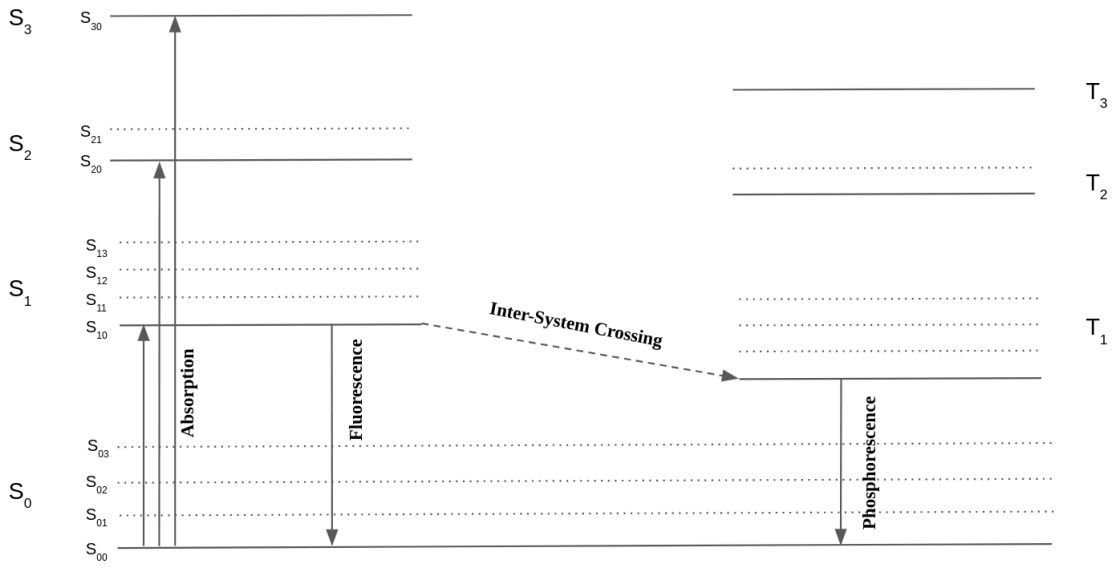

Figure 3: Organic Level Scheme.

which allows for a linear calibration of the long integral into units of keVee.

3 Currently Completed Work

3.1 08/26/24

The current work involved in this prospectus includes an experiment run from 08/19/24 to 08/26/24 populating states in ^9B with the $^7Li(^3He,n)^9B$ reaction. This was achieved with the FN Tandem accelerator at the John D. Fox lab. The ^3He beam has an energy of 7 MeV in the laboratory frame. This was the lowest possible beam energy the Tandem accelerator could coherently transmit the ^3He beam. This energy was minimized to reduce neutron energy so maximum separation could be achieved between different states. The CATRiNA detectors were also placed at backwards angles to further reduce neutron energy. There were three neutron detectors placed at forward angles. All detectors except for the forward detectors were placed at 2 meters from the target ladder. This was done to ensure the anticipated neutron groups would be suitably separated within the ToF spectrum. A beam stop located 3.99 meters after the target ladder was used to halt the beam. The backwards detector angles were 97, 101, 102.5, 105, 108.5, 111.5, 123, 127.5, 129.5, 133.5, 136, 136.5, and 140. The forwards detector angles were 47.5 and 50 degrees. One forwards detector and one backwards detector did not function properly and could not be used.

This beam was incident to a target ladder containing three ^7Li targets, one blank target, one ^{12}C target, and one phosphor target. The ^{12}C and ^7Li targets had a thickness of 200 $\mu g/cm^2$. Since ^7Li quickly oxidizes when exposed to air, the target ladder had to be mounted under vacuum. The phosphor target was used to align the beam due with its luminescence in response to incident radiation. The ^{12}C target was used to populate the ground state of ^{14}O via the $^{12}C(^3He,n)^{14}O$ reaction. This resulted in a single neutron group being visible the resulting excitation spectrum. The presence of this peak allowed for corrections to be applied to the angular displacements and distances of the CATRiNA detectors. The Q-value of this reaction is -1.15 MeV, which can be seen as a sharp peak in the figure below. The sharpness of this peak results from the ground state of ^{14}O having a half-life of 70.606 s, which results in a narrow resonance width. Using this correction, information regarding the populated states in ^9B where extracted. The plots regarding the $^{12}C(^3He,n)^{14}O$ reaction can be seen below.

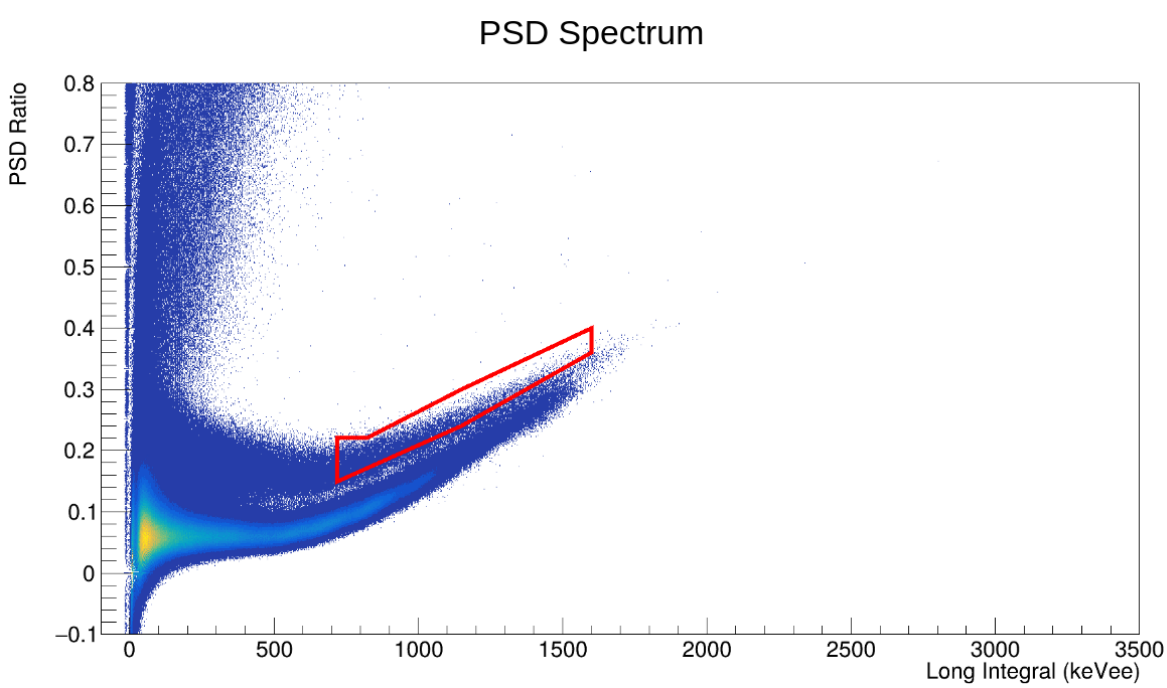

Figure 4: PSD spectrum.

The ToF, neutron energy, Q-value, and excitation spectra for the $^7Li(^3He,n)^9B$ reaction can be seen below. For the ToF spectra, there are two pairs of peaks visible within the spectra. These peaks arise from gamma radiation from the beam exciting the target or the beam stop. These peaks were identified by using the phosphor and the blank targets. The beam only interacts with the phosphor

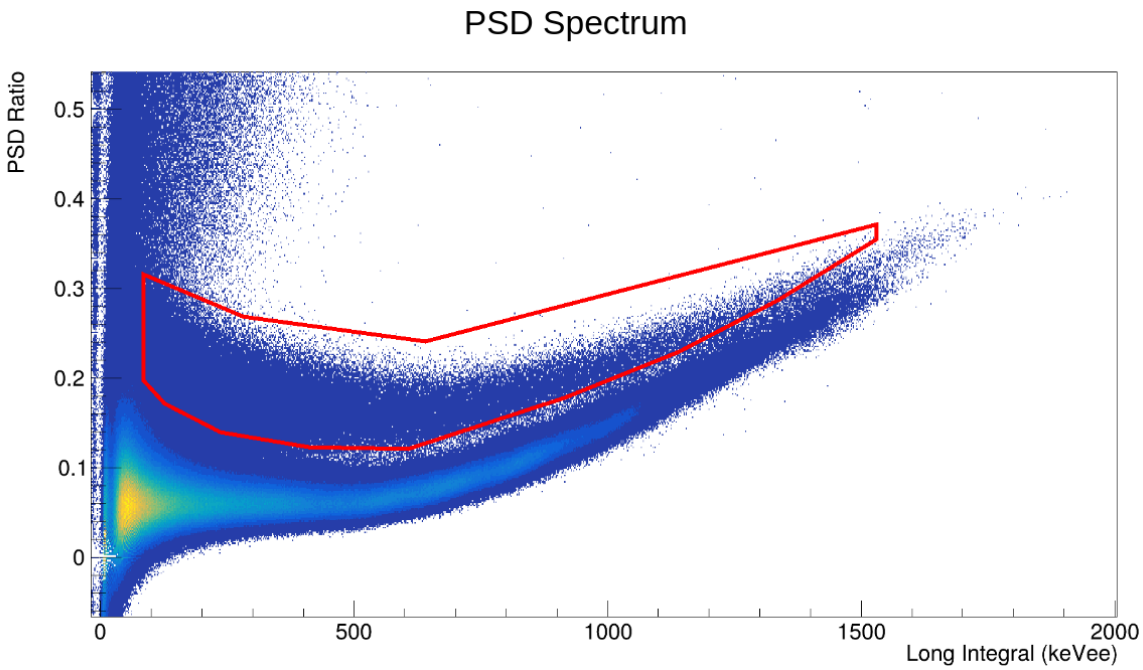

Figure 5: PSD spectrum.

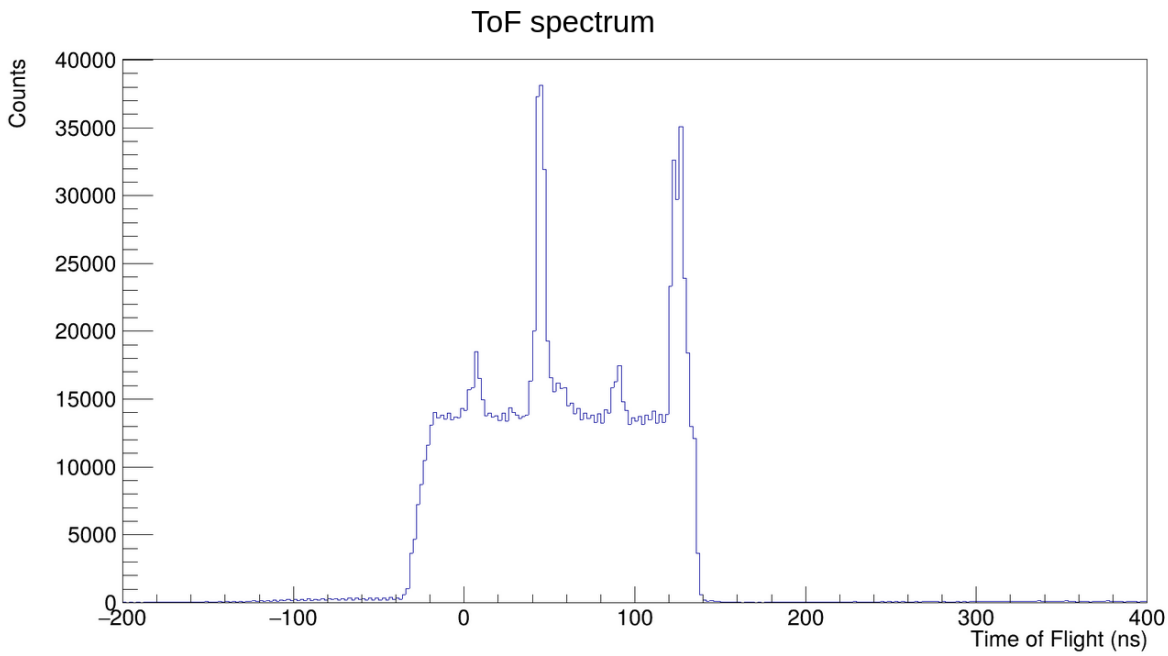

Figure 6: ToF spectrum for the $^{12}C(^{3}He,n)^{14}O$ reaction observed by a CATRiNA detector at 133.5.

target via luminescence. The blank target allows the beam to pass freely to the beam stop. This results in predicable pairs of peaks in each ToF spectrum that can be compared to the $^{7}Li(^{3}He,n)^{9}B$ spectrum as a means of peak identification.

To separate the neutron events from the γ events, gates were applied to the PSD spectrum. The PSD spectra are all calibrated in units of keVee. Two threshold values were applied to the PSD spectra. Following previous work with CATRiNA, a threshold of 85 and 55 keVee was applied to the large and small detectors respectively. This was compared to a uniform threshold of 720 keVee. For the low thresholds, the neutron-gated ToF spectrum becomes highly populated by neutron peaks. There are

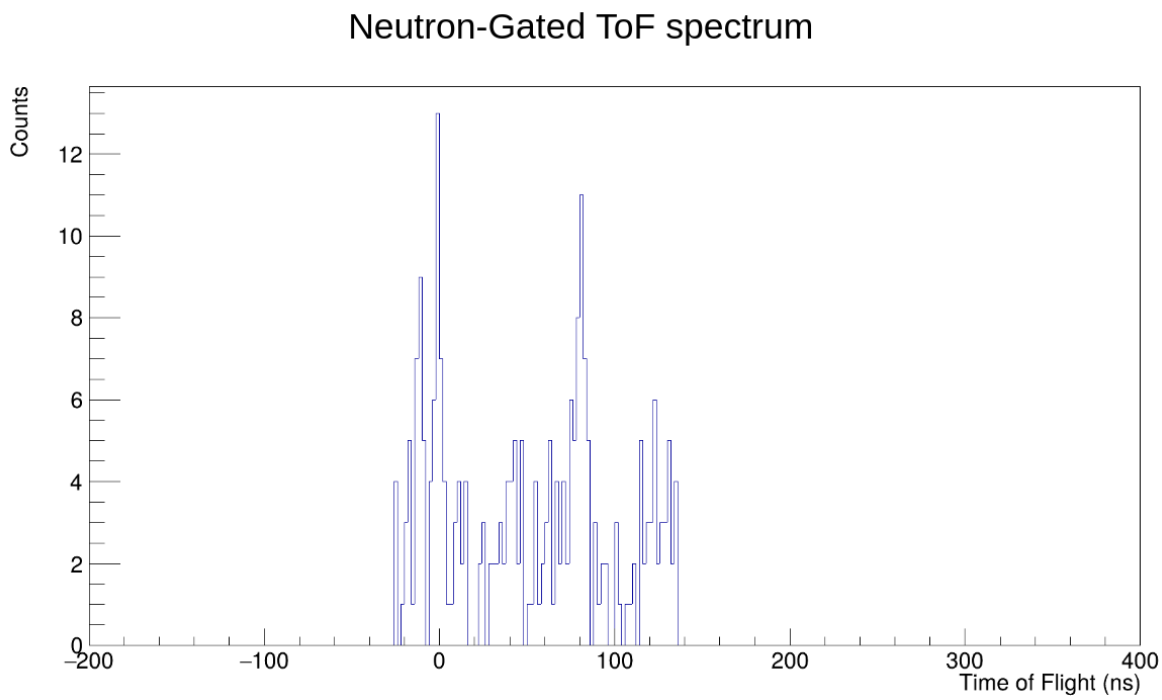

Figure 7: ToFg spectrum.

two prominent neutron groups that correspond to a collection of various states within this spectrum. Due to the limited timing resolution, more precise detail is not available for our ToF spectra.

Q-value and excitation plots were then extracted from the neutron-gated ToF spectra. Certain peaks corresponding to excited states within 9B are identified. The ground state appears as a prominent peak.

Theoretical angular distributions were provided by potentials used in Hodgson and Siemssen [**?**][**?**]. This approach was meant to mirror the angular distribution obtained by K. Gul.

3.2 –

Goals of experiment.
Experimental setup.
Ultimate results and ongoing analysis.

4 Future Work

4.1 Planned Reactions of Interest

Other A = 9 nuclei with states that can be populated with direct reaction and neutron ejectile.

qval_carbon.png

Figure 8: Q-value spectrum.

carbon_excitation_2.png

Figure 9: Q-value spectrum.

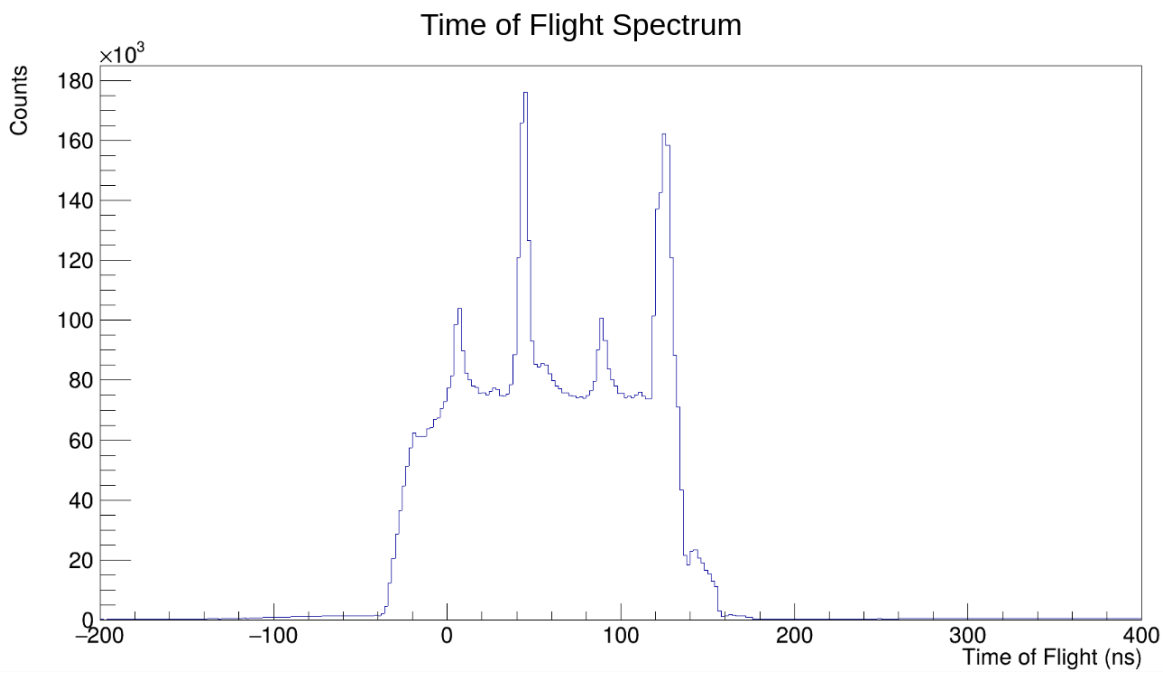

Figure 10: ToF spectrum for the $^{7}Li(^{3}He, n)^{9}B$ reaction observed by a CATRiNA detector.